Abstract

We study Solomonoff axiom induction, a Solomonoff-style prior over axiom systems. A universal machine reads a random tape and emits first-order sentences; the prior probability of sentences labelled true or false is the probability that the emitted theory is consistent and proves every datum. Its predictions form a belief function with three outcomes: proved, refuted, open. We develop its basic theory and relate it exactly to Solomonoff operator induction (SOI): version by version (coin-free, one-tape, Kolmogorov), SAI agrees with SOI over consistent labellers up to constant factors, and on labelled atoms of a fresh predicate it is SOI on strings. Logic adds a lottery: data on which the one-tape inducer has bounded loss although no consistent r.e. theory proves them; in arithmetic, adversarially chosen true data can concentrate the limit posterior on false theories. For Boolean laws and bounded reasoners, the least length of a usable written theory depends exponentially on the pair (reasoner, vocabulary), which program length cannot see. On literal data with charged deduction, the classical reply to Hempel's theoretician's dilemma becomes a theorem about Craig's construction, not about theoretical vocabulary. Under SAI's pricing an approximately true law is a law with a calibrated error model; Kepler's laws are worked through. The upshot: the idealised formalism cannot see the difference between a theory and a program; where shape matters, it is relative to a reasoner and a vocabulary. Separate instances of Claude checked the proofs and computations; no human has yet checked all of the mathematics.

References